

Q-I: Answer precisely in the space given. No steps or explanations required.
[30 min; $10 \times 3 = 30$ points]

1. A set of $\mathcal{X} = \{x_i\}$ of N samples $x_i \in \mathbb{R}^d$, come from a multivariate Gaussian. Covariance matrix of $\mathcal{X}$ is $\Sigma = ?$ (write below the expression for computing covariance matrix)
2. A set of data samples $\mathcal{X} = \{x_i\}$ in $\mathbb{R}^2$ (i.e., $x = [x^1, x^2]^T$) to be linearly transformed to a new set $x' = \{x'_i\}$ also in $\mathbb{R}^2$, such that the first feature/attribute is the sum of two old features/attributes (i.e., $x^1 + x^2$) and second one is the difference of the two old (i.e., $x^1 - x^2$). If the transformation is $x' = Wx$, write the matrix W .
3. A set of data samples $\mathcal{X} = \{x_i\}$ in $\mathbb{R}^2$ has mean $[1, 1]^T$. We need to make all the samples unit norm (L2) and also mean zero (i.e., centered). Write python code snippet to this transformation (use numpy if required).
4. We use KNN for a binary classification problem. “If we increase K , the decision boundary is becoming more and more smooth” Is this statement “TRUE” or “FALSE”?
5. You know the MSE loss popular in linear regression as

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (y_i - w^T x_i)^2.$$

Write the gradient descent based update equation for obtaining w^{k+1} from w^k .

6. We are interested in finding the optimal solution to an overdetermined system of homogenous equations $Ax = 0$, where A is $m \times n$, $b \in \mathbb{R}^m$, and $m > n$. The solution is defined as: (choose the right option from below)
 - (a) Eigen vector corresponding to the largest eigen value of AA^T .
 - (b) Eigen vector corresponding to the smallest eigen value of AA^T .
 - (c) Eigen vector corresponding to the largest eigen value of $A^T A$.
 - (d) Eigen vector corresponding to the smallest eigen value of $A^T A$.

7. Consider a simple diagonal matrix A with all diagonal elements a , i.e., $A = aI$. What could be the SVD (i.e., UDV^T) of A ? (Guess. Use intuition and not by numerically computing).
8. Consider a simple 1D dataset (vector of values) $\{(+1, +), (+2, +), (+3, -), (+4, -)\}$. What will be the SVM decision rule and which are the support vectors?
9. We have one dimensional samples each from two normal distributions $\mathcal{N}(20, \sigma^2)$ and $\mathcal{N}(30, \sigma^2)$. The optimal Bayesian classifier/threshold will be:
- (a) closer to 20
 - (b) closer to 30
 - (c) 25, independently of σ^2
 - (d) more than 30
- (choose the right option)
10. (a) Write the objective function that K-Means algorithm minimizes. (b) Is this algorithm guaranteeing global unique solution (i.e, is this a convex optimization)?